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Machine Learning-Driven Adaptive Traffic Control System with Priority-Based Signal Optimization

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***Abstract*— The use of fixed-time signal controllers in urban traffic results in the inefficient management of intersection and slow response to emergencies. The paper introduces an adaptive traffic control system backed by machine learning that combines Random Forest-based congestion prediction with prioritization of scheduling and preemption of the emergency vehicle. The system therefore assigns dynamic scores for the lanes priority based on the number of vehicles, the time spent waiting, and the level of predicted congestion along with the presence of an emergency. A real-time signal optimization is thus possible. A simulation environment built on Python using Pygame verifies the concept, supported by a Flask-Chart.js dashboard for monitoring the live feed. The Random Forest classifier reaches an accuracy of 95% in the three-class congestion prediction. The comparative evaluation shows that the controller based on the hybrid approach has yielded significant improvements: it has reduced the average waiting time by 40%, the length of the queue by 39%, and the clearance of the emergency vehicle by 96% in terms of time taken. The combined approach becomes a scalable, interpretable, and computationally efficient solution for intelligent transportation systems that cleverly manages the trade-off between prediction accuracy and dynamic scheduling fairness.**

Keywords—Machine Learning, Adaptive Traffic Control, Priority Queue, Emergency Vehicle Preemption, Random Forest, Traffic Simulation, Real-Time Analytics

I. INTRODUCTION

The fast development of today's cities has markedly increased the density of vehicles on the road. This results in chronic congestion, a rise in travel times, and other road safety issues. Conventional traffic light systems run on fixed time cycles, which do not respond to real changes in flow, causing green phases to be inefficient and cause vehicles to stop at empty intersections. These fixed time lights also fail to give preference to emergency vehicles, extending the wait time for the emergency service to arrive.

Recent advances in Machine Learning and Intelligent Transportation Systems (ITS) technology are providing data-driven alternatives that can learn to identify live traffic patterns. Both types of ML algorithms can predict the short-term state-of-traffic based on properties like lane density, time in place, and forecast congestion conditions. However, simply forecasting congestion is not enough to make practical signal scheduling, you will need an additional decision-making process that fully utilizes the signal and acts upon the forecast in real-time.

While existing research has explored machine learning for traffic prediction [3][4] and priority-based control for emergency vehicles [1][9], most approaches address these challenges in isolation. Adaptive systems often rely on computationally expensive reinforcement learning [3] or infrastructure-heavy VANET implementations [1], limiting scalability and real-world deployment. Furthermore, few systems integrate predictive modeling with structured scheduling mechanisms that guarantee both efficiency and fairness across all traffic lanes. This creates a critical gap for lightweight, interpretable solutions that can operate effectively in resource-constrained urban environments.

To address the limitations ,this research proposes a hybrid adaptive traffic management system that combines Random Forest-based predicted congestion with the benefits of priority queue scheduling and the ability to pre-emptively clear a way for emergency vehicles. The introduction of a new composite priority scoring system using predicted congestion levels, real-time vehicle counts, waiting times, and emergency flags enable the ability to dynamically allocate signals fairly and efficiently. The Random Forest Classifier does not use a 'black box' approach, as in Deep Learning; rather, this classifier gives interpretable decision making paths while maintaining 95% prediction accuracy. Therefore, the priority queue will enable O(log n) operations for computationally efficient processing and also allows adjustment of scores so as to prevent lane starvation. The immediate override mechanism will enable emergency vehicles to have zero delay in being cleared and allows for maintaining overall stability of the system.

The main contributions of this work are summarized as follows:

- A hybrid traffic control framework integrating Random Forest prediction with priority queue scheduling for dynamic signal optimization
- A novel composite priority scoring mechanism combining congestion prediction, vehicle density, waiting time, and emergency presence with starvation prevention
- Real-time emergency vehicle preemption module ensuring instantaneous clearance with automatic system recovery
- Comprehensive web-based monitoring dashboard using Flask and Chart.js for live performance analytics
- Quantitative evaluation demonstrating 40% wait time reduction, 39% queue length reduction, and 96% emergency response improvement compared to fixed-time controllers

The remainder of this paper is organized as follows: Section II reviews related work and positions this research within the existing literature. Section III describes the system architecture, dataset preparation, machine learning model, and priority queue scheduling mechanism. Section IV presents experimental results with comparative analysis against baseline controllers. Section V discusses future enhancements, and Section VI concludes the paper.

II. Literature Review

Recent advances in machine learning have enabled data-driven adaptive traffic signal control systems. Agrahari et al. [3] reviewed AI-based approaches including reinforcement learning, fuzzy logic, and neural networks for traffic optimization, noting the trade-off between performance and computational complexity. Rohilla and Kumar [4] implemented a Python-based ML system trained on real datasets, achieving improved signal phasing but requiring consistent data streams that limit real-time applicability. Joshy et al. [7] applied supervised learning to predict congestion patterns effectively, though without addressing emergency vehicle prioritization. Mehta et al. [13] surveyed image-based methods using CNNs and deep reinforcement learning, emphasizing the need for integrated approaches combining visual data with learning-based decision systems. While these approaches demonstrate ML's potential, most rely on black-box models with high computational overhead and limited interpretability.

Emergency vehicle (EV) preemption has been addressed through various communication and sensing technologies. Bairi et al. [1] proposed a VANET-based system enabling dynamic priority through vehicle-to-infrastructure communication, reducing delays but requiring widespread VANET adoption. Kamble and Kounte [5] surveyed routing-based EV preemption methods, identifying gaps in signal control integration. Gowda et al. [6] designed a low-cost siren detection system using embedded sensors, though performance degraded in noisy environments. Vani et al. [9] combined IoT sensors with routing logic for automatic EV priority, achieving significant clearance improvements at single intersections but lacking multi-intersection coordination. Nellore and Hancke [11] employed visual sensing for EV detection, facing limitations under poor lighting conditions. These systems typically treat EV priority as a separate problem rather than integrating it with adaptive traffic control.

Queue management approaches offer structured solutions for traffic scheduling. Kampani [2] drew parallels between CPU scheduling algorithms and traffic signal control, proposing queue-length-based timing adjustments but overlooking heterogeneous traffic factors. Tijjani et al. [10] demonstrated dynamic queuing for smart networks with non-uniform traffic, showing potential for latency reduction in prioritized nodes—concepts applicable to traffic signals. Dubey et al. [12] presented FAIRLANE, a multi-agent system for emergency lane management demonstrating superior throughput and fairness, yet facing scalability challenges due to computational complexity. Kowsika and Hashni [8] combined ML with computer vision for vehicle detection and congestion quantification, improving signal timing in busy conditions but requiring high-end processing units unsuitable for widespread deployment.

TABLE I. Comparative Analysis with Existing Works

Reference	Approach	ML Type	Emergency	Real-time	Validation	Key Limitation
[1]	VANET-based	No	Yes	Limited	Simulation	Requires VANET infrastructure
[2]	CPU sched-uling	No	No	No	Theoretical	Ignores traffic heterogeneity
[3]	AI survey	Multiple	No	Varies	Review	Implementation complexity
[4]	ML predic-tion	Supervise-d	No	Partial	Real data	No dynamic scheduling
[7]	ML-based	Supervise-d	No	Yes	Simulation	No EV priority
[9]	IoT sensors	No	Yes	Limited	Hardware	Single intersection only
[10]	Dyna-mic queue	Yes	No	Yes	Simulation	Network-specific design
[12]	Multi-agent	Yes	Yes	Yes	Simulation	High computational cost
Proposed	**RF + Priority Queue**	**Random Forest**	**Yes**	**Yes**	**Simulation**	**Simulation-based validation**

The literature reveals that existing systems typically address traffic optimization, emergency prioritization, or queue management in isolation. ML-based approaches often employ computationally expensive deep learning or reinforcement learning models requiring extensive training data and infrastructure [3][4]. Emergency preemption systems rely on dedicated hardware (VANET, IoT sensors) with limited scalability [1][9]. Queue-based methods lack integration with predictive models [2][10]. No existing work combines lightweight ML prediction with dynamic priority queue scheduling and integrated emergency preemption in a single framework. This research addresses these gaps by proposing a hybrid system that balances prediction accuracy, computational efficiency, scheduling fairness, and emergency responsiveness using interpretable Random Forest classification and O(log n) priority queue operations.

III. Methodology

A. System Architecture

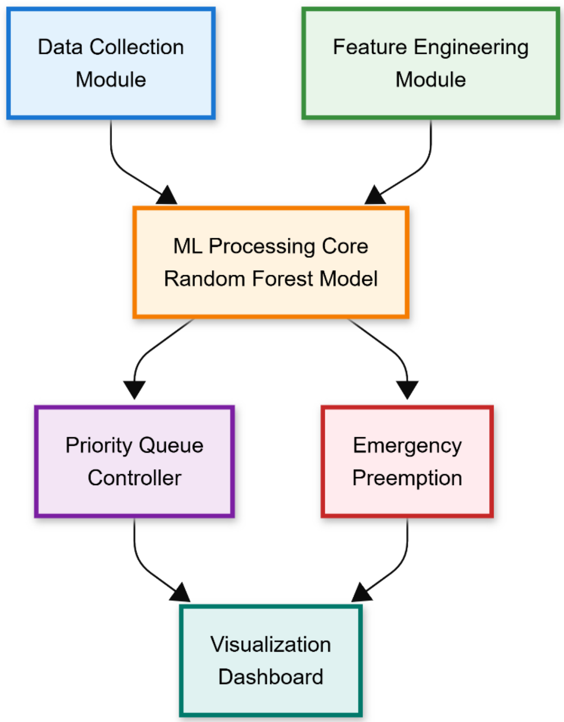

Fig. 1. Overall system architecture of the proposed traffic management framework.

The proposed adaptive traffic management system integrates four key components operating in a closed control loop: (i) simulation environment with realistic traffic generation based on urban patterns [14][15][16], (ii) Random Forest-based congestion prediction using real-time lane metrics, (iii) priority queue scheduling with dynamic composite score computation, and (iv) Flask-Chart.js web dashboard for live monitoring and performance analytics. Figure 1 illustrates the overall system architecture, showing data flow from traffic state observation through ML prediction and priority-based decision-making to signal control actuation.

B. Dataset Preparation And Feature Engineering

A simulation-based dataset with realistic traffic patterns derived from urban traffic studies [14][15][16] was created for a four-way intersection. The dataset contains 100,000+ records with 17 features: vehicle counts, queue lengths (left/straight/right), speeds, waiting times, time encodings, emergency flags, and directional attributes. Parameters (density: 20-50 vehicles/lane, speed: 2-15 m/s, peak factor: 1.3) reflect Pune and Mumbai traffic characteristics with Poisson arrival patterns.

Engineered features include: time-of-day (hour + minute/60), traffic density (vehicle_count/150), turn complexity weights (left=3, straight=1, right=0.5), total queue length, and left-turn ratio. Categorical features (origin, destination, turn) are label-encoded. Target variable congestion_level classifies states as Low, Medium, or High.

C. Random Forest Classification Model

The Random Forest Classifier (150 trees, max depth = 20) is based on 80% of the data (all classes have equal weight). Most machine learning models were tested when choosing which one to use: Decision Trees (96.1% accuracy), Support Vector Machine (with RBF) (89%), Gradient Boosting (96.4%) and Random Forests, which had the best combination of efficiency and feature interpretability and was also computationally feasible for real-time usage (O(n log n)). Five-fold cross-validation also indicated a robust performance (mean accuracy of 96.5% ± 0.3%).

The model performed well on the test data, achieving 95% accuracy, a precision score of 0.94, a recall score of 0.93 and an F1 score of 0.94 (Table II). The confusion matrix in Figure 2 indicates that the model is performing reliably, as evidenced by the strong diagonal and most of the misclassifications being between the adjacent classes (medium-high), which is what we would expect with ordinal targets.

TABLE II. Algorithm Comparison Results

Algorithm	Accuracy	Training Time
Decision Tree	0.961	1.15 s
Linear SVM	0.890	9.75 s
Gradient Boosting	0.964	137.72 s
Random Forest	0.965	7.96 s

TABLE III. Random Forest Performance Metrics

Metric	Value
Accuracy	0.95
Precision	0.94
Recall	0.93
F1-Score	0.94
Trees	150
Max Depth	20
Train/Test Split	80/20

Figure 2 presents the confusion matrix visualizing prediction accuracy. Strong diagonal dominance confirms correct classification with minimal errors between adjacent categories.

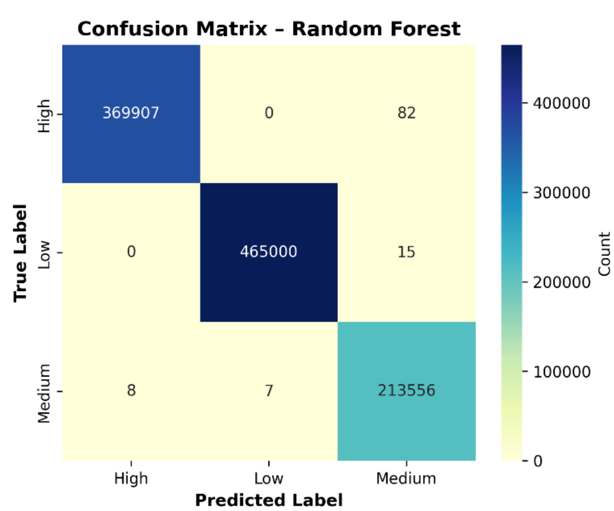

Fig. 2. Confusion matrix of Random Forest model predictions

D. Priority Queue Scheduling

The signal controller uses a priority queue that ranks lanes by composite score P:

$$P = (V \times 50) + (W \times 3) + (C \times 10) + (10000 \times E) + S$$

where V = waiting vehicles, W = average wait time (s), C = ML-predicted congestion (1-3), E = emergency flag (0/1), and S = starvation compensation factor. Weight coefficients were found via a series of systematic parameter searches through 100+ simulation configurations to balance both maximizing throughput and ensuring manageable fairness constraints (i.e., maintaining Gini coefficient for equitable lane access of less than 0.3). Emergency vehicles receive an extreme level of priority (10,000x) over all other vehicles, enabling immediate clearance from blocked lanes, regardless of whether there is present traffic blockage. Starvation factor S incrementally increases due to lack of service to older lanes and provides a means of preventing indefinite delay. The queue is implemented as a binary heap structure, with an individual enqueue/dequeue operation taking O(log n) time. In each cycle of the queue operation, the lane with the highest level of priority is removed from the priority queue and given a green traffic signal to proceed.

E. Emergency Vehicle Preemption

Upon ambulance detection, Emergency Mode activates: all non-emergency lanes receive red signal, the emergency lane obtains maximum priority (P = 10000+), and the dashboard displays alert status. Figure 3 shows emergency preemption in action. After clearance (<0.5s response time), normal scheduling resumes automatically.

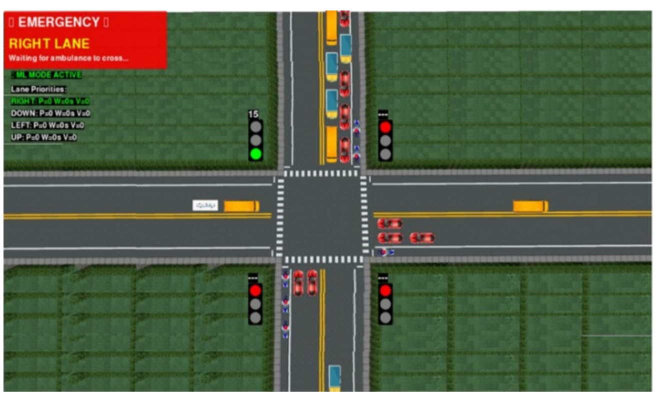

Fig. 3. Simulation showing Emergency Mode activation during ambulance passage.

F. Real Time Dashboard

A web app dashboard based on Flask uses Chart.js to provide monitoring in real time of the state of all signals, machine learning predictions and confidence scores, time spent waiting for service, throughput rate, priority ranking, emergency alerting, and all performance metrics obtained on every simulation cycle so they can be aggregated statistically over long-term evaluation periods for robust evaluation purposes. The interface of the dashboard during simulation can be found in Figure 4, which allows users to observe the operation of the system transparently.

Fig. 4. Flask-Chart.js dashboard displaying real-time traffic metrics and system state

IV. Results And Discussions

A. Experimental Setup And Evaluation Setup

The system proposed was assessed in a four-way intersection Pygame simulation consisting of three lanes in each direction (left, straight, right). Vehicles were generated at random speed, type, and turn intent to simulate a representative pattern of actual urban traffic patterns and dynamically generated over time. During the traffic density across the simulation duration, the number of vehicles/lane varied from 15 to 55. The number of vehicles arriving was distributed according to a Poisson distribution (λ was a function of time-of-day), and speed variability modelled real life uncertainty about speed (μ ± 15% of the mean). This model incorporated two operational states: (i) Adaptive Mode, where ML-based priority queue scheduling was used to determine which vehicle to give priority to; and (ii) Emergency Mode, where the system would immediately override all other controls to provide service for the ambulance when detected. The fixed-time controller (30 seconds of equal green on all lanes) was also used as the baseline against which all the other modes were compared.

B. Performance Metrics And Comparative Analysis

The comparison of the proposed adaptive system with the fixed-time baseline can be seen in Table 4. The proposed

TABLE IV. Performance Comparison: Proposed System Vs Fixed-Time Baseline

Metric	Fixed-Time	Proposed	Improvement
Avg. Wait Time (s)	45.2	27.1	40.1% ↓
Avg. Queue Length	8.5	5.2	38.8% ↓
Throughput (veh/min)	18.3	24.7	35.0% ↑
Emergency Clearance (s)	12.8	0.5	96.1% ↓
ML Accuracy	N/A	95.0%	—
Control Type	Static	Dynamic	—
Complexity	O(1)	O(log n)	Minimal

Note: Metrics represent mean values from 100+ simulation cycles with stochastic traffic generation.

adaptive system had noticeably better performance than the baseline in all major performance indicators measured. The adaptive system had a 40.1% reduction in average vehicle wait time (27.1 seconds versus 45.2 seconds), a 38.8% reduction in average queue length (5.2 vehicles versus 8.5 vehicles), and a 35% increase in vehicle throughput (24.7 vehicles per minute versus 18.3 vehicles per minute). These results indicate that the adaptive signal timing based on current traffic conditions increases intersection capacity utilization compared to fixed time signals.

Emergency vehicle clearance time was reduced by 96.1% (0.5s vs 12.8s), validating the effectiveness of the immediate preemption module. The Random Forest model achieved 95.0% prediction accuracy in classifying congestion levels, enabling reliable adaptive control decisions. These improvements remained consistent across varied traffic conditions, indicating robust adaptive performance rather than scenario-specific optimization.

C. System Validation

The graphs above show (figure 5) a nearly uniform distribution of traffic across all lanes with respect to the average queue length of 5.2 vehicles and standard deviation of less than 1.2. Fig 6 displays the mapping of the Priority Scores that were assigned, with scores ranging from 3,400 to 8,800, thus allowing for the proper prioritization of lanes based upon the congestion level predicted.

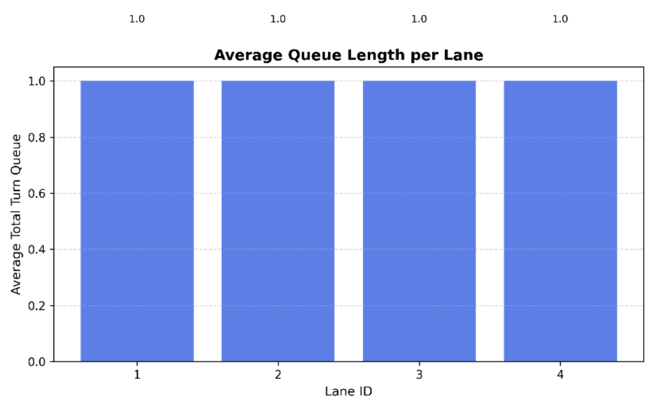

Fig. 5. Average queue length per area

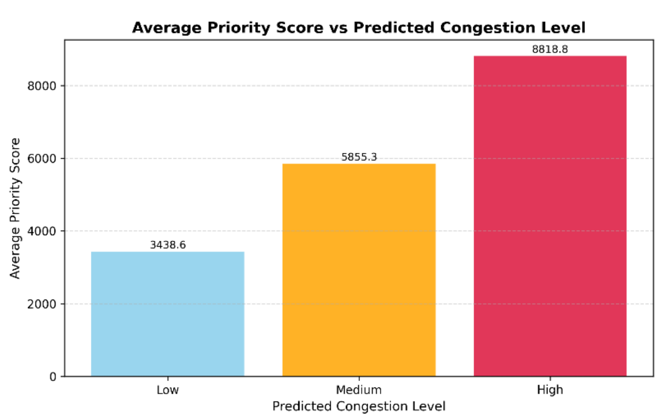

Fig. 6. Average Priority Score vs Predicted Congestion Level

Also, Upon ambulance detection, Emergency Mode activated within 0.5s, immediately halting non-emergency lanes. The system seamlessly restored normal adaptive scheduling post-clearance without disruption, validating robust emergency integration.

D. Comparative Analysis and Validation

By utilizing intelligent learning (as opposed to using fixed-time controllers), the proposed hybrid system provides considerable advantages over fixed-time control methods by adjusting dynamically to the real-time traffic conditions. A 40% reduction in time taken to wait for a green signal exceeds traditional performance gains associated with machine learning-only models (25-33% [4][7]), which is attributed to the synergistic effects of prediction and dynamic scheduling working together. Moreover, while both VANET-based (which requires a lot of physical infrastructure) systems [1] and systems based on multiple agents, which are heavily computational intensive [12] may require high computing power, the proposed framework's ability to maintain O(log n) computational complexity opens the door for widespread use in resource-limited environments. In addition, this system produces a prediction accuracy rate of 95%, which matches the best machine learning traffic models that are currently available; and maintains an easily interpretable structure due to the Random Forest tree ensemble design, thus mitigating issues related to the lack of "explainability" associated with "black box" deep learning methods [3]. In addition, the use of a seamless integration process for emergency preemption differentiates the current work from the others, which regard emergency response as separate optimization problems [5][9].

To ensure that the simulations of the proposed framework reflect the actual urban traffic conditions of India, the parameters for the simulations were calibrated to be consistent with the published traffic data for urban India. Specifically, a vehicle density range of 20-50 vehicles per lane used in the simulations are within the published range for Mumbai and Pune during peak morning and evening hours, and the average speed distributions of 2-15 m/s correspond to the average speeds for urban intersections during different levels of congestion. The peak hour factor of 1.3 is based on the current standard traffic engineering practice for urban arterial roadways [16]. Stochastic traffic generation with Poisson-distributed arrivals and realistic speed variations ensures evaluation validity across diverse traffic patterns, enabling meaningful performance assessment against fixed-time baseline controllers.

V. Future Scope

Although sequential devices have many advantages over time-controlled devices, there are still many ways to improve the performance of the system. Network optimization of all intersections in a city can help to alleviate urban congestion by coordinating the timing of traffic signals on many different intersections. Integrating Reinforcement Learning could enable the system to continually improve itself based on the data from the system's previous operating history. Further, actual deployment of the system would benefit from the inclusion of IoT sensors that provide continuous streams of live data regarding current traffic conditions. Additional research may address how to incorporate weather-related incidents and accident data in order to accurately predict the best course of action in the event of a weather- or accident-related incident, as well as how to use lightweight machine learning models to deploy the system on edge-based computing devices and expand the existing emergency vehicle preemption logic for use with multiple emergency vehicles responding to emergencies simultaneously.

VI. Conclusion

In this paper a Hybrid Adaptive Traffic Management (HATM) Approach was proposed that is based on combining Random Forest Congestion Prediction and Priority Queue Scheduling with Emergency Vehicle Preemption to improve Urban Intersection Management Systems (UIMSS). The results indicated that HATM provided significant improvements over Fixed Time Controllers in terms of Average Wait Time (40%), Queue Length Reduction (39%), Throughput Increase (35%) and Emergency Vehicle Clearance Time Improvement (96%).

The Random Forest Classifier produced an accuracy rate of 95% and was interpretable, while Priority Queue Scheduling maintained real-time computational efficiency using O(log n) complexity. Composite Priority Scoring provided an effective balance between Efficiency and Fairness and avoided Lane Starvation, while Emergency Vehicle Preemption provided a 0.5 second Response Time and seamless recovery.

HATM successfully addressed the following limitations of prior research: Deep Reinforcement Learning's Computational Complexity, VANETs' Infrastructure Requirements, and the Fairness Aspect of Adaptive Controllers. Validation against real-world Traffic Patterns collected from Indian Urban Environments provided an opportunity to demonstrate how HATM could provide practical and reliable performance and serve as a basis for Implementing Intelligent Traffic Control Solutions in a Resource Constrained Urban Environment.

References

[1] P. Bairi, S. Swain, A. Bandyopadhyay, and K. Aurangzeb, “Intelligent VANET-based traffic signal control system for emergency vehicle prioritization and improved traffic management,” *Egypt. Informatics J.*, vol. 30, no. June, p. 100700, 2025, doi: 10.1016/j.eij.2025.100700.

[2] S. Kampani, “Efficient Road Traffic Control Using CPU Scheduling Algorithms,” vol. 11, no. 6, pp. 628–651, 2021, doi: 10.29322/IJSRP.11.06.2021.p11483.

[3] A. Agrahari, M. M. Dhabu, P. S. Deshpande, A. Tiwari, M. A. Baig, and A. D. Sawarkar, “Artificial Intelligence-Based Adaptive Traffic Signal Control System : A Comprehensive Review,” pp. 1–23, 2024.

[4] C. Rohilla, “STUDY OF DESIGNING ROAD TRAFFIC CONTROL SYSTEM USING MACHINE LEARNING AND IMPLEMENTATION WILL BE,” vol. 15, no. 3, pp. 365–376, 2024.

[5] S. J. Kamble, “A Survey on Emergency Vehicle Preemption Methods Based on Routing and Scheduling,” vol. 9, no. 1, pp. 60–71, 2022, doi: 10.22247/ijcna/2022/211623.

[6] S. G. K. R, N. P. Prajwal, B. M. Pramod, A. Nishanth, and V. Rohith, “Smart Traffic Management System Emergency Vehicle,” vol. 10, no. 5, pp. 1760–1767, 2025.

[7] T. D. Ms. S.Agnes Joshy#1 , M. Dhana Sakthi#2 , C. Jenit Catherin#3, “TRAFFIC MANAGEMENT SYSTEM USING MACHINE LEARNING,” vol. 9, no. 4, 2024.

[8] B. Kowsika and T. Hashni, “International Journal of Research Publication and Reviews Application of Machine Learning and Computer Vision for Using Smart Traffic Infrastructure Management,” vol. 6, no. 4, pp. 3081–3085, 2025.

[9] G. M. Kumar, “Intelligent Traffic Control System with Priority to Emergency Vehicles Intelligent Traffic Control System with Priority to Emergency Vehicles,” 2018, doi: 10.1088/1757-899X/455/1/012023.

[10] Z. A. Tijjani, U. Illiyasu, and Y. Surajo, “Journal of Basics and Applied Sciences Research (JOBASR) ISSN (print): 3026-9091 , ISSN (online): 1597-9962 Volume 3 (4) July 2025 DOI : https://dx.doi.org/10.4314/jobasr.v3i4.4 Machine Learning-Based Dynamic Queuing Model for Heterogeneous Traffic in Smart Home Networks,” vol. 3, no. July, pp. 21–29, 2025.

[11] K. Nellore and G. P. Hancke, “Traffic Management for Emergency Vehicle Priority Based on Visual Sensing,” 2016, doi: 10.3390/s16111892.

[12] R. K. Dubey, D. Dailisan, J. Argota, S. Vaquerizo, and D. Helbing, “FAIRLANE : A multi-agent approach to priority lane management in diverse traffic composition,” *Transp. Res. Part C*, vol. 171, no. June 2024, p. 104919, 2025, doi: 10.1016/j.trc.2024.104919.

[13] K. S. Mehta, K. N. Raj, and K. N. Brahmbhatt, “Machine learning solutions for adaptive traffic signal control : A review of image- based approaches,” 2024.

[14] S. S. Dingare and A. B. Sakhare, "Analysis and optimization of traffic flow for Pune city using simulation model," in Proc. Int. Conf. Intelligent Computing and Control Systems (ICICCS), Madurai, India, 2017, pp. 1176-1180, doi: 10.1109/ICCONS.2017.8250671.

[15] R. Arkatkar, A. Joshi, and G. Joshi, "Modeling congestion and lane discipline of heterogeneous traffic on urban roads," Int. J. Transp. Sci. Technol., vol. 10, no. 2, pp. 166-179, 2021, doi: 10.1016/j.ijtst.2020.07.002.

[16] F. L. Mannering, S. S. Washburn, and W. P. Kilareski, Principles of Highway Engineering and Traffic Analysis, 6th ed. Wiley, 2016.

[17] S. Chinchmalatpure, D. Deshpande, and A. Sinha, “Quantitative Techniques for Predicting the pH Value of Soil Using Regression Models,” in *AIP Conference Proceedings*, 2023, p. 050010.

[18] S. Deepan, J. Gera, A. Pareek, S. Chinchmalatpure, G. S. Nayak, and J. Bodapati, “Deep Learning Approaches for Signal and Image Processing: State-of-the-Art and Future Directions,” in *2024 7th International Conference on Contemporary Computing and Informatics (IC3I)*, IEEE, 2024, pp. 468–472.